



Unit 4 · Outcome 1 · Climate Change

One molecule, four clocks

KK05

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 05 Year 11 & 12 revision One molecule, four clocks A single carbon-dioxide molecule in the air above you is part of a story that runs on four different clocks at once — a clock that ticks every season , one that ticks every year , one that ticks across centuries , and one that ticks across millennia

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



01 The same record, four zoom levels

01 The same record, four zoom levels Think of a digital map. Zoom right in and you see one street; zoom out and you see a city; zoom further and you see a whole continent. Nothing about the land changed — only your zoom level did, and each zoom reveals a pattern the others hide. Greenhouse-gas concentrations work exactly the same way. The dot point gives you the four "zoom levels" explicitly, so learn them as a set. ✧ Anchor analogy — zoom levels Seasons = the street view (the yearly saw-tooth). Years = the suburb (the steady ~2–3 ppm climb). Centuries

02 Seasons — the planet takes a breath

02 Seasons — the planet takes a breath Why does CO₂ dip and rise every single year? Because most of the world's land — and therefore most of its forests, crops and grasslands — sits in the Northern Hemisphere. When the northern spring arrives, billions of plants leaf out and photosynthesise, pulling CO₂ out of the air faster than it is returned. The global CO₂ level falls. When northern autumn comes, leaves drop and decompose and plants respire more than they fix, so CO₂ rises again. ✧ Anchor analogy — the breathing planet The saw-tooth is the Ear

03 Years — a staircase that never stops climbing

03 Years — a staircase that never stops climbing Strip the saw-tooth away and average each year, and a brutally simple pattern remains: every year is higher than the last. Since the late 1950s the annual average has climbed from about 315 ppm to over 425 ppm. The rate itself has grown — from under 1 ppm/yr in the early record to about 2–3 ppm/yr today. The year-to-year climb isn't perfectly smooth. Natural events nudge it: El Niño years Bigger jumps Warmer, drier tropics mean stressed forests draw down less CO₂ and release more through fire and decay, s

04 Centuries — the hockey stick

04 Centuries — the hockey stick Zoom out to centuries and the saw-tooth vanishes into a single bold shape. For most of the last 10,000 years CO₂ hovered near 280 ppm. Then, from roughly 1750 — the start of the Industrial Revolution — the line bends sharply upward and keeps going. The long flat handle and the sudden blade give the curve its nickname: the hockey stick. This is the time scale where the human signal becomes undeniable. The rise tracks the growth of coal, then oil and gas; it accelerates after 1950 alongside population, industrialisation a

05 Millennia — the ice-core time machine

05 Millennia — the ice-core time machine How do we know what the air was like before 1958 — before any instrument existed? We read ice cores. As snow falls on Antarctica and Greenland and compresses into ice, it traps tiny bubbles of the actual atmosphere. Drill down through the layers and each bubble is a literal sample of ancient air; the deeper you go, the older it is. The longest cores stretch back 800,000 years. ✧ Anchor analogy — Earth's diary An ice core is Earth's diary, written one snowfall at a time and never edited. Each trapped bubble is a

06 The four human sources you must name

06 The four human sources you must name The dot point names four human activities explicitly. Examiners expect you to identify each one and the gas/mechanism behind it — not just "humans pollute." Learn this as a four-square. Source 1 Burning fossil fuels Coal, oil and gas are ancient carbon locked underground. Combustion oxidises that carbon, releasing CO₂ that the slow carbon cycle had removed over millions of years. The single largest source. → CO₂ Source 2 Manufacturing cement Making cement "cooks" limestone (CaCO₃) to make lime (CaO). The reacti

07 An Australian lens — Cape Grim

07 An Australian lens — Cape Grim You don't have to fly to Hawaii to see this. On the wild north-west tip of Tasmania sits the Cape Grim Baseline Air Pollution Station, run by CSIRO and the Bureau of Meteorology since 1976. Positioned to sample the cleanest air on Earth as it sweeps in off the Southern Ocean on the "roaring forties" westerlies, Cape Grim is the Southern Hemisphere's answer to Mauna Loa. Comparing the two records makes a beautiful teaching point about the seasonal clock : ~6–7 ppm Seasonal saw-tooth amplitude at Mauna Loa (Northern Hemis

08 Exam craft — command words & structure

◆ COMMAND WORDS

08 Exam craft — command words & structure Marks in this dot point are lost less on knowledge and more on matching the answer to the command word. Decode the verb first. Identify / State Name it, no explanation. "Photosynthesis lowers CO₂ in the growing season." One line. Describe Say what happens — the pattern and its features. The saw-tooth falls in NH summer, rises in NH winter, ~6–7 ppm each year. Explain Give the cause / mechanism — why it happens. Photosynthesis exceeds respiration in the growing season, so CO₂ is drawn down. Distinguish / Compar

09 Practice questions

★ PRACTICE & SELF-MARK

09 Practice questions Five VCAA-style questions, 18 marks total, spanning the command words above. Try each in your head or on paper first, then open the model. Each shows a weak response, an annotated strong response, and a marking guide — then mark yourself. Your score builds in the dock at the corner. Q1 The graph of atmospheric CO₂ measured at Mauna Loa shows a small saw-tooth pattern that repeats every year. Explain what causes this seasonal pattern. Seasons Explain 2 marks ► Reveal model responses & marking guide X Weak response — 0/2 The CO₂ go

10 Recap & checklist

◆ RECAP / CHEAT SHEET

10 Recap & checklist The one-screen summary Seasons → small annual saw-tooth (~6–7 ppm) → natural, NH photosynthesis vs respiration. Years → steady ~2–3 ppm/yr climb, wobbled by El Niño/La Niña → mostly human. Centuries → "hockey stick" from 280→430 ppm since ~1750 → human (fossil fuels, cement, agriculture, land use). Millennia → ice-core cycles 180↔280 ppm every ~100,000 yr → natural (Milankovitch). Today is far outside this envelope. Four human sources: fossil-fuel burning, cement (calcination), agriculture (CH₄, N₂O), land-use change (released

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

The CO₂ goes up and down because of the seasons changing through the year. [Restates the pattern; gives no mechanism — no marks.]

✓ STRONG / MODEL ANSWER

In the Northern-Hemisphere spring and summer, the many land plants there grow rapidly, so photosynthesis removes CO₂ from the air faster than respiration and decomposition return it, lowering the concentration. In autumn and winter plant growth slows and decay and respiration release CO₂ faster than it is taken up, raising it again — producing the annual saw-tooth.

✗ WEAK ANSWER

The seasonal change happens every year and the long-term trend is over a long time. The seasonal one is natural and the long-term one is caused by humans. [Touches cause but ignores the patterns; "happens every year / over a long time" is not a real description — only the cause contrast scores.]

✓ STRONG / MODEL ANSWER

Seasonal variation: a repeating saw-tooth of about 6–7 ppm that falls and rises within each year and nets to zero; it is natural, caused by Northern-Hemisphere plant photosynthesis exceeding respiration in the growing season. Long-term trend: a sustained rise of roughly 280 to 430 ppm that does not reverse; it is human-driven, caused mainly by fossil-fuel combustion, cement, agriculture and land clearing adding CO₂ faster than natural sinks remove it.

✗ WEAK ANSWER

The student is wrong because humans are causing climate change by burning fossil fuels. CO₂ is higher now. [Asserts a conclusion but barely uses the ice-core evidence the question demands, and never weighs the claim — "evaluate" needs both sides.]

✓ STRONG / MODEL ANSWER

The claim has a grain of truth: CO₂ has genuinely cycled naturally between ~180 and ~280 ppm over the 800,000-year record, driven by Milankovitch orbital cycles. However the ice cores actually refute the claim: today's ~430 ppm sits far above the entire natural envelope (which never exceeded ~300 ppm); the change has occurred in about two centuries rather than tens of thousands of years; and the driver is human emissions, not orbital change. On balance the evidence shows the current rise is not just another natural cycle — it is larger, faster and outside the natural range.

✗ WEAK ANSWER

Cutting down trees and farming. Trees store carbon and farming makes gases. [Two valid activities identified (2 marks), but the "how" is too vague to score the describe marks — which gas? what mechanism?]

✓ STRONG / MODEL ANSWER

. Cement manufacture: heating limestone (CaCO_3) to make lime drives off CO_2 in the calcination reaction itself, releasing carbon independently of the fuel used.
2. Agriculture: livestock release methane (CH_4) through enteric fermentation and flooded rice paddies emit CH_4 , while nitrogen fertilisers release nitrous oxide (N_2O) — both potent greenhouse gases. (Land-use change would also be accepted: clearing forests releases stored carbon and removes a sink.)

✗ WEAK ANSWER

Cape Grim is in Australia and Mauna Loa is in Hawaii so they get different readings because they are in different places. [Notes a difference in location but gives no reason tied to vegetation or mixing — "different places" is not a mechanism.]

✓ STRONG / MODEL ANSWER

The seasonal swing is driven by land-plant photosynthesis, and the Southern Hemisphere has far less vegetated land (mostly ocean), so its seasonal drawdown is much weaker — giving Cape Grim a smaller saw-tooth than Northern-Hemisphere Mauna Loa. The long-term rise is the same at both because greenhouse gases mix throughout the whole atmosphere within a year or two, so the human-driven increase is global rather than local.